R: Introduction and review

EC421, Set 01(R)

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Econometrics

Recall: Applied econometrics, data science, analytics require:

1. Intuition for the **theory** behind statistics/econometrics (assumptions, results, strengths, weaknesses).
2. Practical knowledge of how to **apply theoretical methods** to data.
3. Efficient methods for **working with data** (cleaning, aggregating, joining, visualizing).

This course aims to deepen your knowledge in each of these three areas.

- 1: As before.
- 2-3: **R**

R

R

What is R?

To quote the R project website:

> R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS.

R

Why are we using R?

1. R is **free** and **open source** saving both you and the university 💰💵💰.
2. *Related:* Outside of a small group of economists, private- and public-sector **employers favor R** over `Stata` and most competing softwares.
3. R is very **flexible and powerful** adaptable to nearly any task, *e.g.*, metrics, spatial data analysis, machine learning, web scraping, data cleaning, website building, teaching (these slides).

R

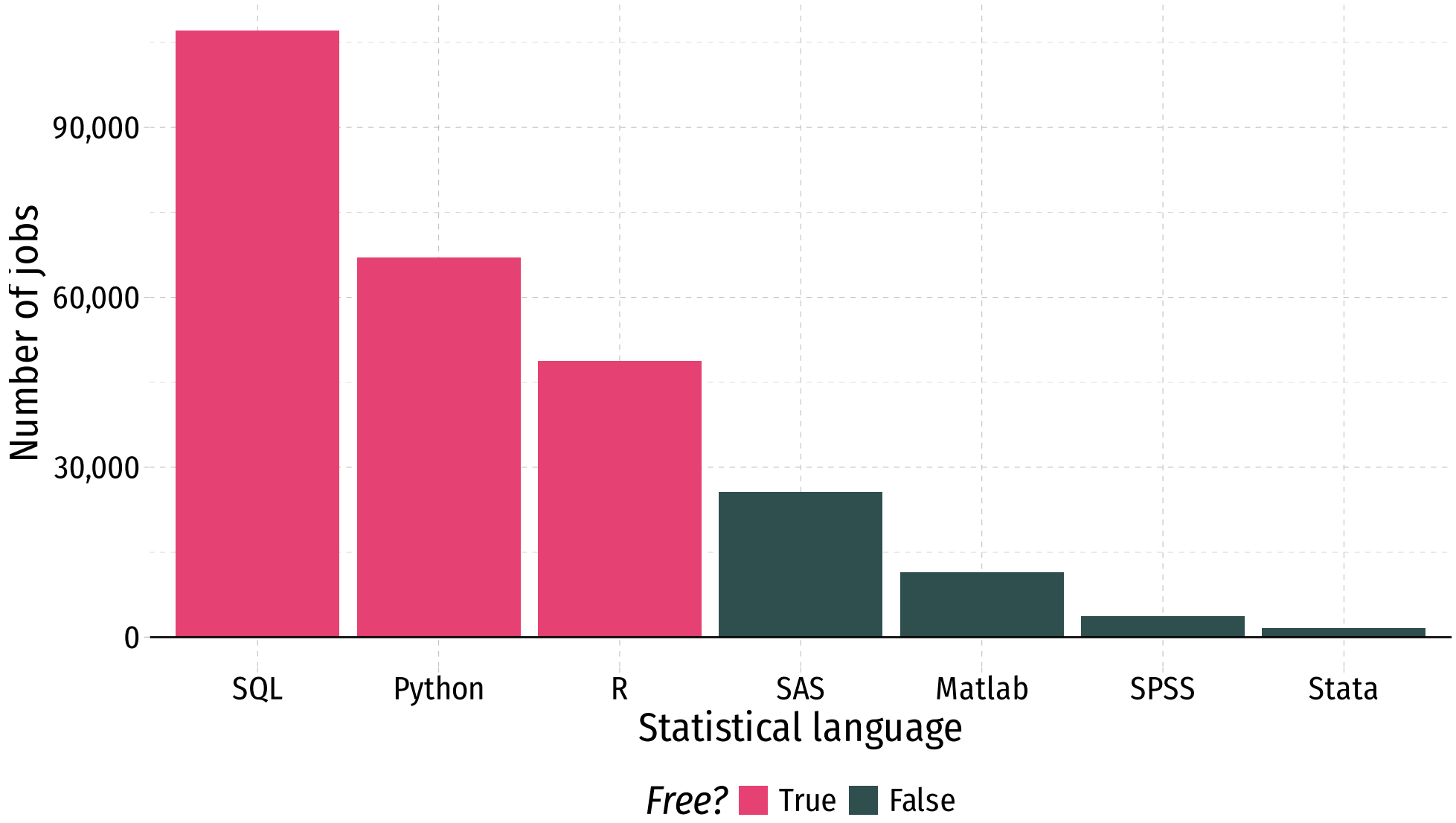
Why are we using R?

4. *Related:* R imposes **no limitations** on your amount of observations, variables, memory, or processing power. (I'm looking at **you**, `Stata`.)
5. If you put in the work,† you will come away with a **valuable and marketable** tool.
6. I 💖 **R**
7. R is a nice gateway to (and plays well with) other programming languages (*e.g.*, Python, SQL, C++, JavaScript).

[†] Learning R definitely requires time and effort.

Comparing statistical languages

Number of job postings on Indeed.com, 2019/01/06



R + Examples

R + Data

```r
# Load the gapminder dataset (via the package)
library(gapminder)
# View the first 5 rows
gapminder |> head(5)
```

```
# A tibble: 5 × 6
  country     continent  year lifeExp      pop gdpPercap
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
1 Afghanistan Asia       1952    28.8  8425333      779.
2 Afghanistan Asia       1957    30.3  9240934      821.
3 Afghanistan Asia       1962    32.0 10267083      853.
4 Afghanistan Asia       1967    34.0 11537966      836.
5 Afghanistan Asia       1972    36.1 13079460      740.
```

R + Regression

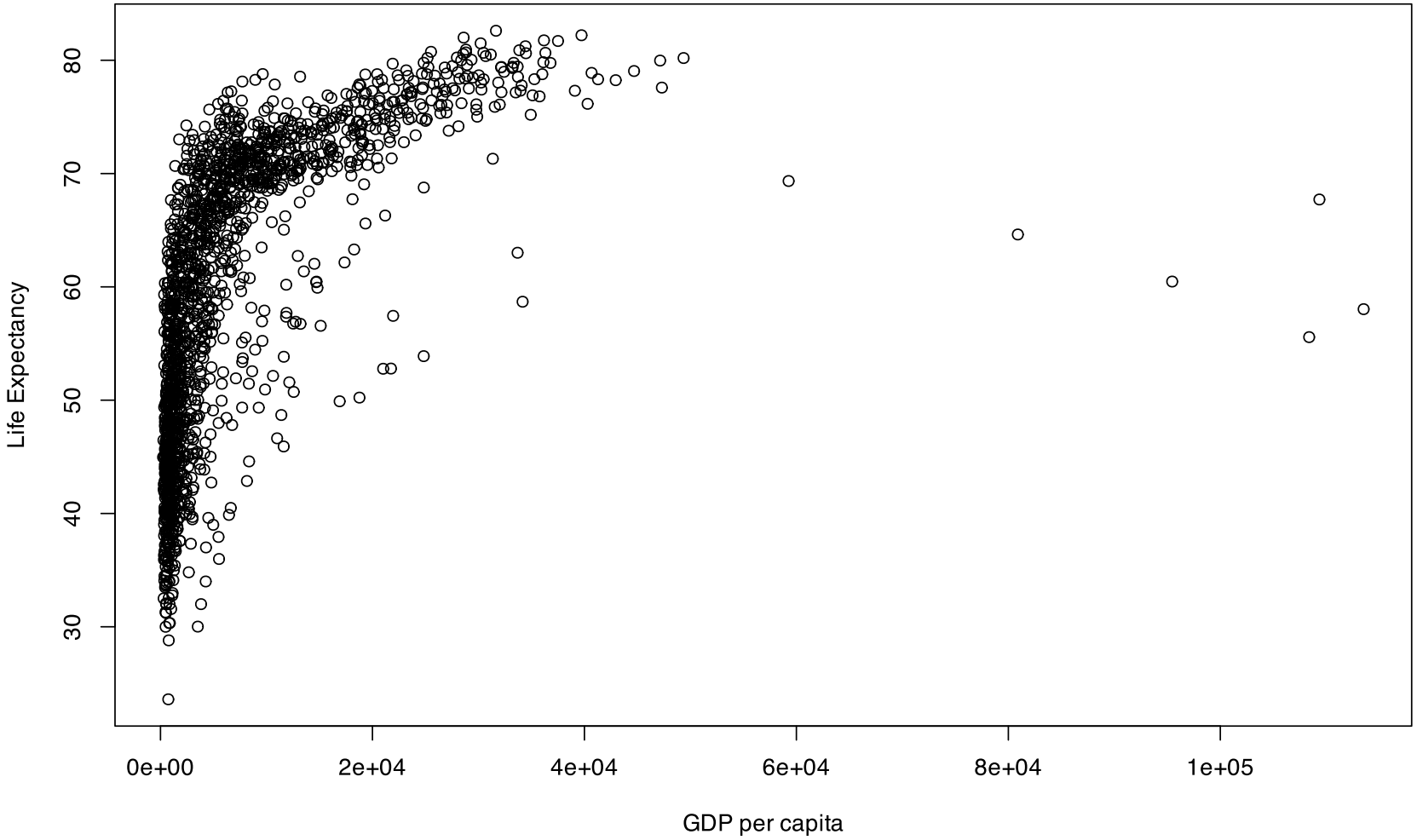
```r
# Fit a linear regression
fit = lm(lifeExp ~ 1 + I(gdpPercap / 1e3) + I(pop / 1e6), data = gapminder)
# Show the results
fit |> summary() |> coef()
```

R + Regression

```r
# Fit a linear regression
fit = lm(lifeExp ~ 1 + I(gdpPercap / 1e3) + I(pop / 1e6), data = gapminder)
# 'tidy' the results (from the broom package)
fit |> broom::tidy()
```

```
# A tibble: 3 × 5
  term              estimate std.error statistic   p.value
  <chr>                <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)       53.6       0.322      166.   0
2 I(gdpPercap/1000)  0.768     0.0257      29.9  4.04e-158
3 I(pop/1e+06)       0.00973   0.00238      4.08 4.72e-  5
```

R + Plotting (w/ `plot`)

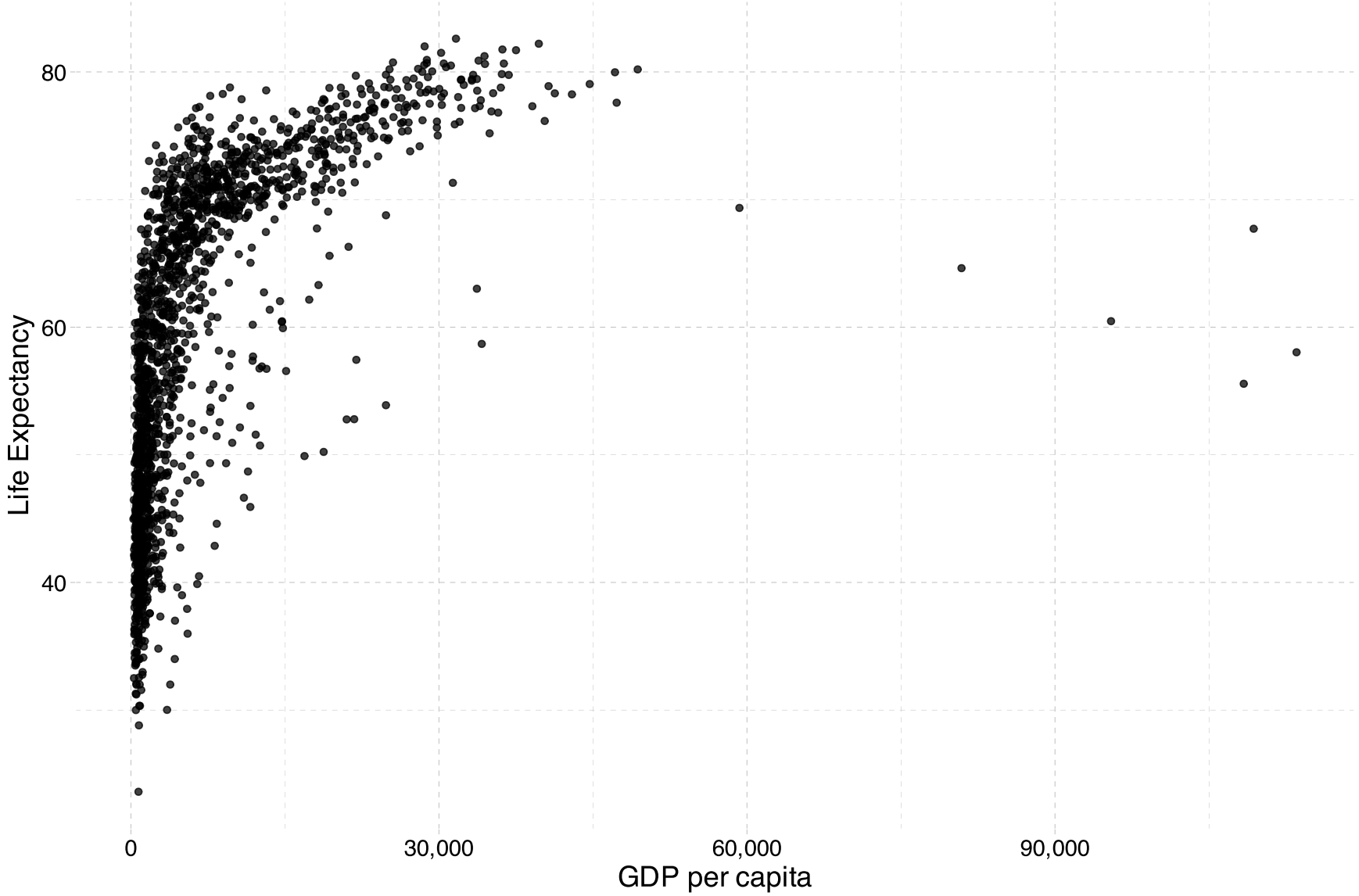


R + Plotting (w/ plot)

```r
# Load packages with dataset
library(gapminder)

# Create simple plot
plot(
  x = gapminder$gdpPercap,
  y = gapminder$lifeExp,
  xlab = 'GDP per capita',
  ylab = 'Life Expectancy'
)
```

R + Plotting (w/ ggplot2)

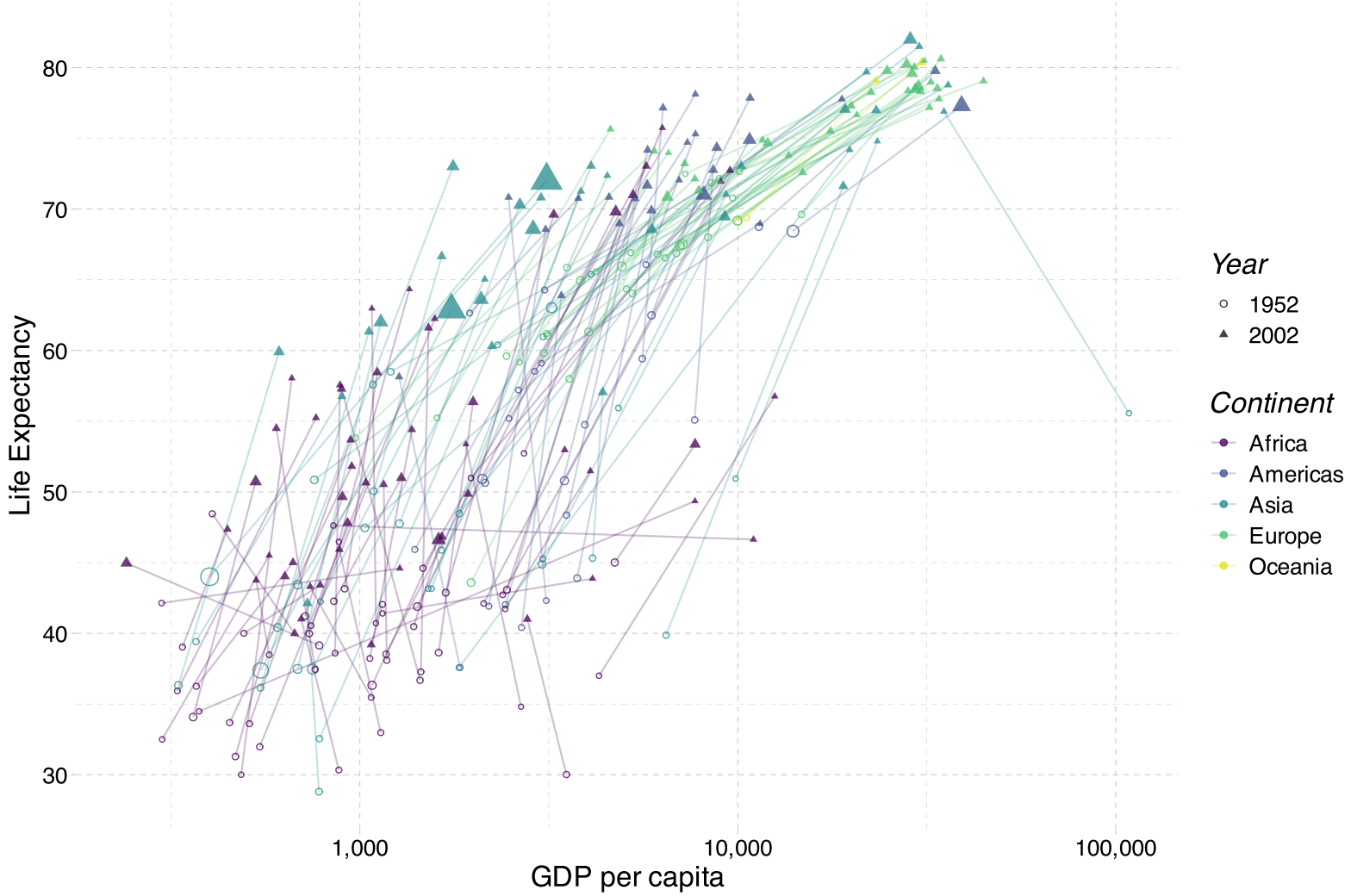


R + Plotting (w/ ggplot2)

```r
# Load packages
library(gapminder)
library(dplyr)

# Plot with ggplot2
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp)) +
  geom_point(alpha = 0.75) +
  scale_x_continuous('GDP per capita', label = scales::comma) +
  labs(y = 'Life Expectancy') +
  theme_pander(base_size = 16)
```

R + More plotting (w/ ggplot2)

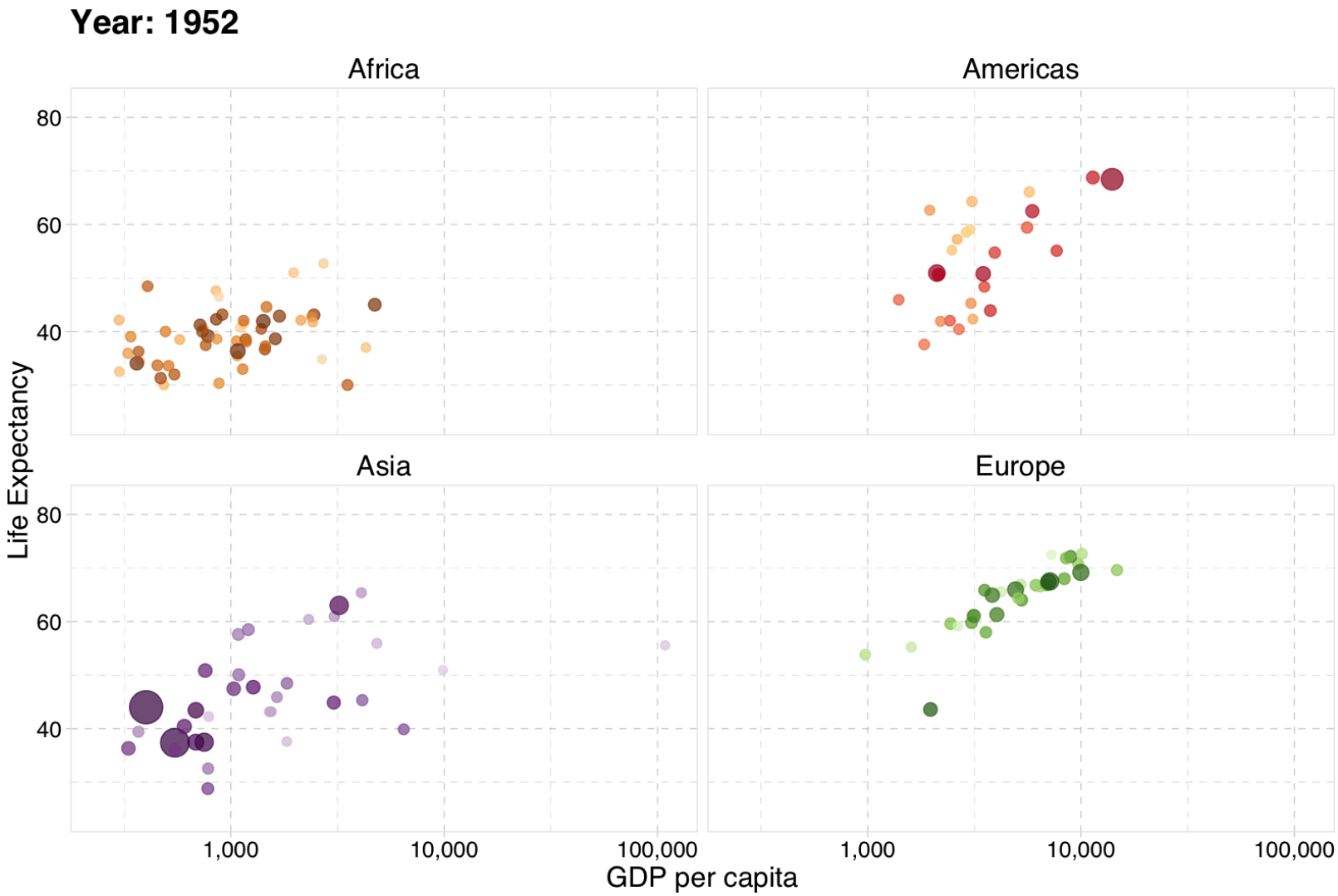


Notice we've moved to *log-10 scale* on the x-axis.

R + More plotting (w/ ggplot2)

```r
# Plot with ggplot2
ggplot(
  data = filter(gapminder, year %in% c(1952, 2002)),
  aes(x = gdpPercap, y = lifeExp, color = continent, group = country)
) +
  geom_path(alpha = 0.25) +
  geom_point(aes(shape = as.character(year), size = pop), alpha = 0.75) +
  scale_x_log10('GDP per capita', label = scales::comma) +
  labs(y = 'Life Expectancy') +
  scale_shape_manual('Year', values = c(1, 17)) +
  scale_color_viridis('Continent', discrete = TRUE, end = 0.95) +
  guides(size = FALSE) +
  theme_pander(base_size = 16)
```

R + Animated plots (w/ gganimate)

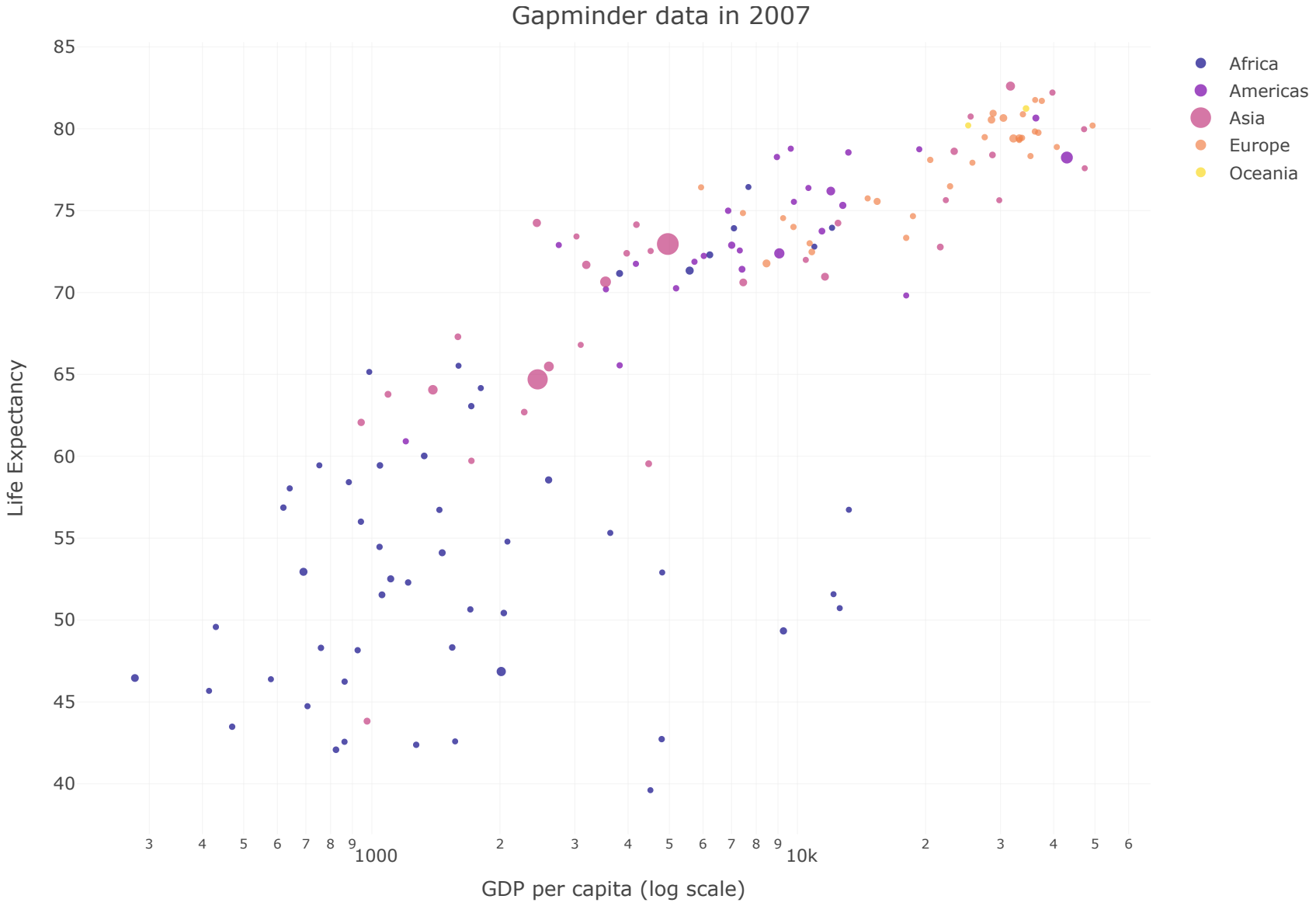


Gapminder

R + Animated plots (w/ gganimate)

```r
library(gganimate)
ggplot(
  data = gapminder |> filter(continent != 'Oceania'),
  aes(gdpPercap, lifeExp, size = pop, color = country)
) +
  geom_point(alpha = 0.7, show.legend = FALSE) +
  scale_colour_manual(values = country_colors) +
  scale_size(range = c(2, 12)) +
  scale_x_log10('GDP per capita', label = scales::comma) +
  facet_wrap(~continent) +
  theme_pander(base_size = 16) +
  theme(panel.border = element_rect(color = 'grey90', fill = NA)) +
  labs(title = 'Year: {frame_time}', y = 'Life Expectancy') +
  transition_time(year) +
  ease_aes('linear')
```

R + Interactive plots (w/ plotly)



```r
plot_ly(
  data = gapminder %>% filter(year == 2007),
  x = ~gdpPercap,
  y = ~lifeExp,
  type = 'scatter',
  mode = 'markers',
  size = ~pop,
  color = ~continent,
  colors = viridis::plasma(n = 5, end = .93),
  text = ~paste(
    'Country: ', country,
    '<br>GDP per capita:', scales::dollar(gdpPercap, 1),
    '<br>Life Expectancy:', scales::comma(lifeExp, 1),
    '<br>Population:', scales::comma(pop)
  )
```

R + Maps

Referer is required by tile usage policy of OpenStreetMap's volunteer-run servers: osm.wiki/Blocked
Access
Referer i
tile usag
OpenStre
voluntee
osm.wiki
OR 99
Franklin Boulevard
Mill Street
Mill Race
Brooklyn Subdivision
East 8th Avenue
East 12th Avenue
East 13th Avenue
East 14th Street
East 15th Avenue
East 17th Avenue
East 18th Avenue
East 19th Avenue
East 20th Avenue
Willamette Street
Olive Street
Oak Street
Pearl Street
Ferry Street
Alder Street
High Street
Hilyard Street
University Street
Agate Street
Moss Street
Villard Street
Orchard Street
Walnut Street
Garden Avenue
OR 99
OR 126 BUS
University of Oregon
Hayward Field
Pioneer Memorial
South Eugene High School
Hendricks Park Rhododendron Garden
Leaflet | © OpenStreetMap, ODbL

Getting started with R

Starting R

Installation

- Install `R`.
- Install `RStudio`.
- Install `Quarto`.
- **Optional/Overkill:** `Git`
  - ↳ Create an account on `GitHub`
  - ↳ Register for a student/educator discount.
  - ↳ For installation guidance and troubleshooting, check out Jenny Bryan's website.
- **Note:** Many UO labs have R installed and ready (helpful in a pinch).

R resources

Free(-ish)

- Time
- Google (inevitably StackOverflow)
- Codex, Claude, Copilot, ...
- Data services at the UO library
- Your classmates
- Your GE and me
- R resources, *e.g.*, here and here
- `swirl` and learnr

Money

Short online courses, *e.g.*, DataCamp

Starting R

Some R basics

You will dive deeper into R in lab, but here are six big points about R:

1. Everything is an **object**. `foo`
2. Every object has a **name** and **value**. `foo = 2`
3. You use **functions** on these objects. `mean(foo)`
4. Functions come in **libraries** (**packages**) `library(dplyr)`
5. R will try to **help** you. `?dplyr`
6. R has its **quirks**. `NA; error; warning`

Next: Metrics review(s)